

SQL Basics - Assignment 01

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Course: Database Management Systems - MySQL Basics

Date Issued: Sep 3, 2025

Due Date: Sep 5, 2025 12:00 PM

Instructions

This assignment is designed to test your understanding of fundamental SQL concepts in MySQL, specifically focusing on `SELECT`, `WHERE`, `DISTINCT`, `ORDER BY`, Comparison Operators, Boolean Operators (`AND`, `OR`, `NOT`), `IN`, `LIKE`, `BETWEEN`, and `NOT NULL`.

Please answer all questions by writing the appropriate SQL queries. Assume you are working with a database for a small library, and the tables provided below are available.

Database Schema

For this assignment, assume the following table structures:

Table: Books

Column Name	Data Type	Constraint	Description
BookID	INT	PRIMARY KEY	Unique identifier for each book
Title	VARCHAR(255)	NOT NULL	Title of the book
AuthorID	INT	FOREIGN KEY (references Authors.AuthorID)	ID of the author
Genre	VARCHAR(100)		Genre of the book
PublicationYear	INT		Year the book was published
AvailableCopies	INT	NOT NULL	Number of copies currently available

Table: Authors

Column Name	Data Type	Constraint	Description
AuthorID	INT	PRIMARY KEY	Unique identifier for each author
FirstName	VARCHAR(100)	NOT NULL	Author's first name
LastName	VARCHAR(100)	NOT NULL	Author's last name
Nationality	VARCHAR(100)		Author's nationality

Questions

Write the SQL query for each of the following:

1. **DDL** to Create the above tables and **DML** to populate the tables
2. **SELECT & WHERE**: Retrieve the **Title** and **PublicationYear** of all books published after 2000.
3. **DISTINCT**: List all unique **Genres** available in the **Books** table.
4. **ORDER BY**: Select the **Title** and **AuthorID** of all books, ordered alphabetically by **Title**.
5. **Comparison Operator**: Find all books with exactly 5 **AvailableCopies**.
6. **Boolean Operator** : Get the **Title** and **Genre** of books published before 1990 AND with the **Genre** 'Fantasy'.
7. **Boolean Operator**: Retrieve the **BookID** and **Title** for books that are either 'Mystery' or 'Thriller' genres.
8. **Boolean Operator**: Select all books that are NOT of the 'Science Fiction' genre.
9. **IN Operator**: Find the **FirstName** and **LastName** of authors whose **Nationality** is 'American' or 'British'.
10. **LIKE Operator**: List the **Title** of all books that start with the letter 'The'.
11. **BETWEEN Operator**: Get the **Title** and **PublicationYear** of books published between 1995 and 2005 (inclusive).
12. **NOT NULL**: Find the **BookID**, **Title**, and **Genre** of all books where the **Genre** is specified (i.e., not NULL).
13. **Combined Operators**: Select the **Title**, **AuthorID**, and **PublicationYear** of books published after 2010, written by authors whose **Nationality** is 'Canadian'

and have more than 2 AvailableCopies. Order the results by PublicationYear in descending order.

Submission Guidelines

- Create a text file named <your_name>_assignment_1.zip.
- Submit your file via ahmed@upsurge.us

Good luck!